

The SharedVehicles Project

The idea behind SharedVehicles is to build a system that's very dynamic, can respond to changes in demand and is cost effective for every one; it has to be a shared system. Of course, owned vehicles will still be a part of the system, but they can't take you far if we've to make the system fast and responsive beyond a point. Consider the following hypothetical situation.

Let's suppose you've got to go from A to D, with the route being A-B-C-D. Imagine that there is a mass transportation system or metro available from B-to-C. If you're at A, to go to B, you can take a 'shared' electric bike, which will be available at A. Then you drop the bike at B, take the metro to C, and you take another electric bike parked at C, to reach D, and leave it there for someone else to pick up.

Of course, in reality, things will be much more complicated than this, and here, I'm simply suggesting one possibility.

If we've to implement such a system, one problem we would face is understanding the requirements of users and the corresponding behaviour of the system. As the system becomes too complex, we may have to employ some powerful concepts from the world of simulation. In such a situation, we can employ agent based modelling and simulation. So what exactly is agent based modelling? How is it useful for urban transportation? And how is it related to the open source movement?

Agent based modelling

All of us have attended physics classes in high school (and for many, in college, too), where the physics teacher started the discussion saying, "Consider a body of mass...". For a physicist, the object of study was a *body*. Similarly, when an artificial intelligence researcher is trying to simulate the behaviour of a person or an intelligent being, he will call either of them an agent. But note that the agent need not be as intelligent as a human—it can be very elementary in terms of behaviour or level of intelligence. Usually, agents have a set of rules governing them, and they behave according to those rules to achieve some objective. Typically, we study the behaviour of an agent in relation to its surroundings or an environment. Sometimes, the agent will be alone, while most of the time agents interact with other agents, and such a setting is called *multi-agent simulation*.

There are many frameworks available to implement such agent based modelling. Many of them are written in Java. For example: Mason - <https://cs.gmu.edu/~eclab/projects/mason/>, Jade: <http://jade.tilab.com/>, etc.

Mason is easy to get started with, to understand, and of course, to manipulate. It has a 'single-process, discrete event simulation library' according to official documentation. It also comes with a visualisation library. Simulation can be run without visualisation, but the latter often helps us to understand simulation better, and can be intelligently adopted to a particular simulation.

Examples of simulation provided in the Mason framework demonstrate the capability of agent based modelling. In one

simulation called *Ant foraging*, the behaviour of ants is modelled. Each ant is an agent that acts on its own; however, it is governed by some rules. Ants discover food, and establish a food trail between the source and the nest. In another simulation called *Virus Infection* we can see the spread of a virus. There are infected individuals or agents, and uninfected individuals or agents, and the simulation studies how the virus spreads. There is also the famous example – John Conway's 'Game of Life'.

It's easy to run Mason. Just run the command `java -jar mason.19.jar` once you've downloaded Mason from the site mentioned above. Mason has very good documentation in terms of a manual - <https://cs.gmu.edu/~eclab/projects/mason/manual.pdf>. The best way to learn Mason is through the examples and tutorials that are provided.

Simulation and the SharedVehicles Project

Now that we've understood the concept of agent based modelling and its possible applications, we can look at how to use it in the SharedVehicles Project. Here passengers, vehicles, buses, the metro, taxis and private vehicles are all *movable*. There can be a *movable* interface or a class. Then we can implement/extend it to give behaviours specific to buses, passengers, metros, etc. We can then go one crucial step further by adding parameters like costs, charge for the electric battery, locations of charging stations, etc, and try to obtain the cost for each person and the profit/loss for different operators. When we reach a point at which we can show that for a specific arrangement, the cost of transportation is much less than what passengers are paying today, and where profits for the operators can be substantial, then we would have achieved a breakthrough. That would create an incentive for the government/entrepreneurs to implement such a system.

To put it in a different way – anybody can contribute ideas in terms of code. Just like any other FOSS project, this will be available to everyone. When the simulation suggests that there can be a better, faster and more efficient system that can be implemented, which can also become profitable, that system will get implemented by a willing entrepreneur.

The SharedVehicles project is available at <https://github.com/project-vidyut/>, where you can also find links to the resources needed to implement the project. You can visit Project Vidyut site: <http://www.projectvidyut.org/> This is a new experiment in which we're trying to solve a burning real-world problem with ideas and experience from the FOSS movement. I request you to contribute in terms of code, ideas, and constructive criticism. 

By: Mahesha Hiremath

The author is an independent data analytics consultant from Bengaluru. He is building a platform for electric vehicles called Project Vidyut, a not-for-profit initiative for promoting the use of electric vehicles, and also to build open source electric vehicles. He can be reached at mahesha-at-projectvidyut-dot-org and tweets with handle @MaheshaHiremath.